

# Asjad Nirban

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## EDUCATION

### University of Virginia

Master's in Computer Science, GPA: 3.81/4.0

Charlottesville, VA

Aug 2024 – Dec 2025

### Mumbai University

B.E. in Computer Engineering, GPA: 3.73/4.0

Mumbai, India

Jul 2019 – May 2023

## EXPERIENCE

### Data Engineering Research Assistant

Darden School of Business, University of Virginia

Charlottesville, VA

Mar 2025 – Present

Tech Stack: LLM, RAG, HPC, SLURM, Databricks, Azure Synapse, Selenium, Linux, Bash Scripting

- Designed and automated scalable data pipelines processing 10TB+ of raw data, improving workflow efficiency by 70%
- Redesigned ETL jobs and scheduling logic, cutting runtime by 60% and improving reliability across research projects
- Integrated fine-tuned LLM models using OpenAI API into data pipelines to parse unstructured filings into structured records
- Utilized MPI and HPC to scale computational tasks across Multi-GPU and Multi-Processing leveraging CUDA cores

### Full Stack Software Developer Intern

Lab Systems (I) Pvt. Ltd.

Mumbai, India

Jun 2023 – Jul 2024

Tech Stack: Solidity, NextJS, MongoDB, GraphQL, NodeJS, Python, Apache eCharts, JIRA, Mocha

- Built full-stack forensic dashboards using ReactJS and Flask, supporting blockchain investigations on systems processing transaction of \$20M+ in cryptocurrency network
- Built scalable enterprise-grade UIs in NextJS and 20+ REST APIs supporting large-scale data queries for enterprise clients
- Delivered Python automation and AI driven forensic services that cut forensic review time by ~70% and improved case-handling efficiency
- Integrated Bitcoin and other data nodes into backend services, enabling fast on-chain lookups for investigation teams
- Deployed cryptocurrency full nodes and architected software for on-premises deployments while managing global clients

## PROJECTS

### Autonomous Driving Fault Simulation Platform

Tech Stack: OpenPilot, MetaDrive, ScenarioNet, Python, Linux, Deep Neural Network

- Engineered a large scale autonomous driving simulation backend by integrating 4TB of Waymo trajectory data into MetaDrive via ScenarioNet enabling automated edge case scenario generation
- Executed and analyzed 3000 plus high risk driving simulations uncovering critical failure patterns in OpenPilot control logic and informing reliability focused system improvements

### Sales Intelligence AI Agent

Tech Stack: Python, Transformers, LLM, PyTorch, FAISS, SQLite, Typer

- Implemented a backend service that ingests company data, ranks leads, and generates personalized emails using a 30B-parameter LLM
- Devised modular pipelines for data ingestion, scoring, and generation, automating ~80% of manual prospect research

### Synthetic Blood Cancer Cell Image Generation

Tech Stack: Python, Deep Learning (CNN-based GANs), PyTorch, Matplotlib, NumPy

- Developed custom GAN models (DCGAN, WGAN, LSGAN, CGAN) by improving training stability and tuning loss functions to generate synthetic blood cancer cell images, addressing severe medical data scarcity
- Achieved high-quality images with ~85% similarity to real data, enabling effective dataset expansion for medical research

## SKILLS

**Programming Languages:** Python, C/C++, JavaScript, TypeScript, SQL, NoSQL, GraphQL, Bash

**Frameworks & Libraries:** React/Next.js, Express.js, Laravel, Node.js, Flask, pytest, Mocha, Tailwind CSS

**Development Tools:** Docker, Git, REST APIs, Linux, CI/CD, SLURM, Postman, AWS, Databricks, Azure Synapse, OpenAI API

**Data & Machine Learning:** Pandas, Polars, Transformers, Selenium, Azure Synapse

**Relevant Skills:** Data Structures & Algorithms, OOP, Software Architecture, Operating Systems, Software Engineering, Distributed Systems

## PATENT

### D. Ein Blockchain-basiertes Medizinlogistiksystem (Granted)

Sep 2023

German Patent File No. 20 2023 102 823.3

- Developed a blockchain-based web application for tracking prescription drugs, enabling secure and transparent pharmaceutical supply-chain management
- Improved end-to-end traceability by **80%** and significantly reduced counterfeit drug occurrences in real-world deployments